Sintering

- Sintering is the process of consolidating either a loose aggregate of powder or a green compact of the desired composition under controlled conditions of temperature and time.
- sintering is, "the thermal treatment of a powder or compact at a temperature below the melting point of the main constituent, for the purpose of increasing its strength by bonding together of the particles".
- Sintering operation is generally carried out at temperatures ranging from 0.7-0.9. Tm (Tm is the melting point in kelvin) of the metal or alloy.
- During sintering, the compact is usually heated in a protective atmosphere such as argon or hydrogen.
- The individual powder particles, which were either loose or physically bonded together, are metallurgically bonded to yield a solid structural part with the desired properties.
- Several changes take place during sintering like shrinkage, formation of solid solution and development of the final microstructure.
- In many cases, sintering results in a reduction/elimination of porosity, leading to densification

- In other instances, sintering may not result in any densification and in certain cases sintering can result in the growth of the compact (i.e. swelling).
- During sintering, the green compact is consolidated and takes on its final shape and microstructure.
- Sintering involves a number of processes, which may take place one after other or may occur simultaneously.
- It is very difficult to clearly define and analyze the sintering process, in spite of a large amount of research efforts.
- There are many conflicting theories and data, and sintering is still an active field of research.
- The latest developments in sintering can be had from the regular conferences devoted to the

TYPES OF SINTERING

Solid State Sintering

This is a commonly occurring process of consolidation of metal and alloy powders (loose powders or compacts).

In solid state sintering, the densification is a result of atomic diffusion in solid state.

Liquid Phase Sintering

- One of the most commonly used methods of achieving rapid densification
- Densification is enhanced by employing a small amount of liquid phase (typically 1 to 10 vol. %).
- The liquid phase co- existing with the powder particles at the sintering temperature has some solubility for the solid.
- The composition of the powder compact and the sintering temperature are chosen such that sufficient amount of liquid is formed between the solid particles of the compact.
- On cooling, the liquid crystallizes or forms a solid phase at grain boundaries binding the grains.
- The application of temperature initially results in some degree of densification due to solid state sintering.
- Once a part of the material melts, the liquid phase formed wets the solid by spreading around the solid particles.

- During this stage, there is a rapid rearrangement of the solid particles leading to increase in density.
- The liquid dissolves the solid until it is saturated.
- If the temperature is maintained above the liquid formation temperature, the second stage of sintering takes place, during which the smaller particles are dissolved and reprecipitated on the larger grains along with an increase in density.
- In the last stage—known as solid phase sintering— grain coarsening takes place and the densification rate slows down.
- This technique is useful for sintering of systems like tungsten-copper and copper-tin.
- The liquid phase sintering also allows sintering of covalent compounds like silicon nitride and silicon carbide, which are otherwise difficult to sinter.
- This technique is also used for the manufacturing of cemented carbide tools where tungsten carbide particles are bonded together by cobalt.

- The densification can also be enhanced by the simultaneous

Activated Sintering

- The addition of a small amount of an alloying element called 'doping' in certain metals and alloy systems increases the rate of densification by as much as 100 times compared to the undoped compacts.
- The best known example of activated sintering is the doping of nickel in tungsten compacts.
- Nickel, which is added as a solution of its salts, gets reduced during sintering and forms a coating of a few monolayer on the surface of the tungsten particles.
- By diffusing over the tungsten particles, nickel accumulates at the necks of the particles from where the nickel penetrates the grain boundaries.
- This presence of nickel at the grain boundaries greatly increases the grain boundary self-diffusion of tungsten.
- Thus the segregation of the additive at the interparticle boundary creates a short circuit

- To promote activated sintering, any additive must form a low-melting phase during sintering and also must segregate at the grain boundaries.
- The additive must also have a very high solubility for the base metal so as to promote rapid diffusion of base metal through the additive phase.

- Other examples of activated sintering includes the passing of a shock wave through a powder so as to increase the defects in the structure resulting in increased dislocation motion, as well as by the use of non-stoichiometry or by

Reaction Sintering

- Reaction sintering is a special form of rapid sintering technique to process high temperature materials, resulting from chemical reaction between the individual constituents, giving very good bonding.
- Reaction sintering takes place when two or more components chemically react during the sintering process to form the final product.
- The green compact made up of a mixture of powders is heated to a temperature at which the individual constituents will react to form the final product.
- Reactive sintering may involve a transient liquid phase also.
- Reactive sintering is nearly spontaneous once the

- Reactive sintering has been successfully used to process various aluminide type intermetallic compounds.
- A typical example is the reaction between alumina and titania to form aluminium titanate at 1553 K which then sinters to form a densified product.
- Reaction sintering may also involve heating in a gas atmosphere, which reacts with the powder compact resulting in the formation of the desired compound.
- A typical example is the production of reaction bonded silicon nitride made by sintering green compact of silicon powders in nitrogen atmosphere at 1523 to 1673 K for times upto 70 h

Rate Controlled Sintering

- Rate controlled sintering (RCS) refers to the sintering process where the densification rate is controlled rather than the heating rate as in conventional sintering.
- By controlling the rate of densification through all stages of sintering, this process can avoid gas entrapment in pores and rapid grain growth.

Microwave Sintering

- Microwaves have also been used to carry out sintering of powder compacts.
- Microwave sintering has the advantage of producing a finer grain size compared to pressure less sintering due to the faster heating rate and shorter dwell time at the selected temperature.
- However, problems such as hot spots and melting of compact may occur.

Self-Propagating High-Temperature Synthesis

- Self-propagating High-Temperature Synthesis (SHS) is an exothermic combustion process that uses the heat generated during a reaction to sinter the material.
- SHS reactions can take place between gases, liquids and solids.
- Porous materials such as filters and catalyst supports with a higher strength than those obtained by conventional methods have been produced by SHS.
- The process, although is very fast and economical, needs an additional densification

Gas Plasma Sintering

- The sintering source used in Gas Plasma Sintering can be a DC cathode source, radio frequency waves or microwaves.
- The advantage in gas plasma sintering, as in the case of microwave sintering, is the possibility of producing fine grain size material.

Spark Plasma Sintering

- Spark Plasma Sintering (SPS) is a process similar to hot uniaxial pressing in which a momentary DC current pulse is generated in addition to joule heating, by producing a 'spark' between the powder particles.
- The temperatures during SPS can reach up to 5000°C to 10000°C, but only momentarily so that no melting is allowed to take place.
- Functionally graded materials can be produced by this method.

SINTERING THEORY

- Sintering may involve

(i) single component systems (e.g. pure metals and ceramics), wherein shrinkage is a major factor

(ii) multicomponent system, involving more than one phase, where several processes like solid solution formation and liquid phase formation may also occur in addition to densification.

Apart from these, the operating mechanisms also differ significantly: in the case of multicomponent systems, for example, inter-diffusion occurs with the concentration gradient being a major driving force for sintering in addition to the self- diffusion caused by surface tension and capillary forces.

This is in contrast to sintering of single component system, where self-diffusion is the major material transport mechanism and the driving force resulting from a chemical potential gradient due to surface tension and capillary forces between particles

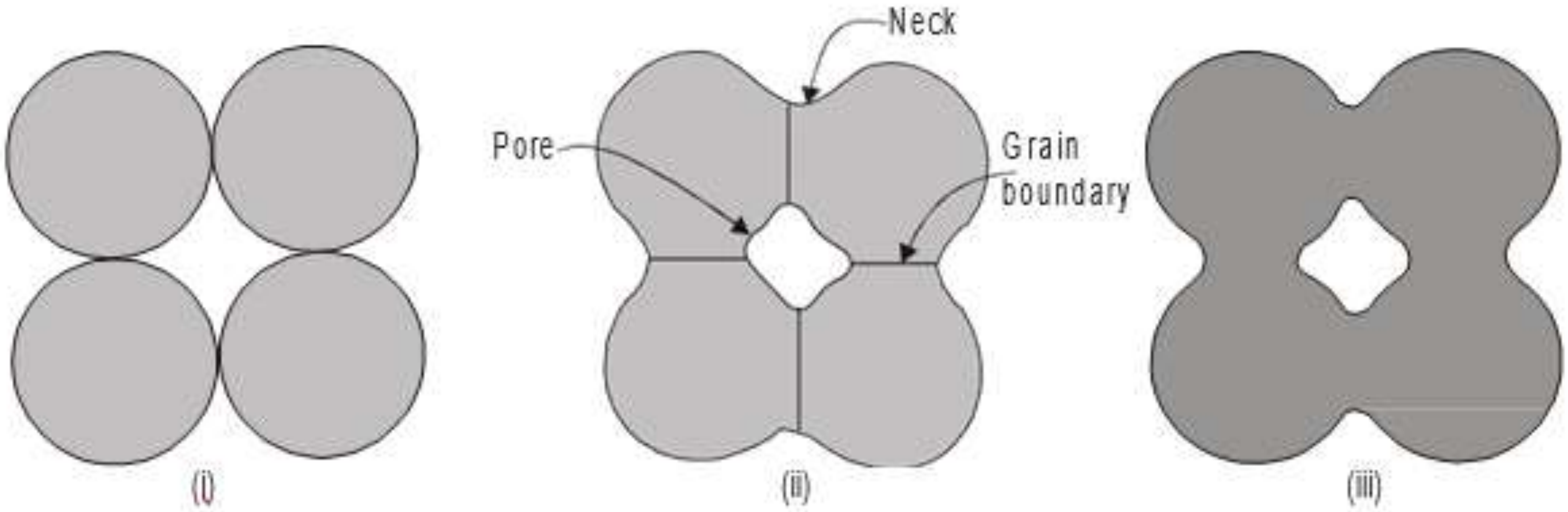
- One of the first theories on sintering was proposed by Sauerwald in 1922.
- According to this theory, there were two stages in sintering namely, adhesion and recrystallisation.
- Adhesion occurs during heating due to atomic attraction while recrystallization, occurring at temperatures above 0.7 Tm, consists of diffusion of atoms between adjacent grains.
- Recrystallization includes changes in the microstructure, phase changes, grain growth, precipitation and pore shrinkage.
- Later, it was recognized that sintering proceeded by particle rounding and neck formation and that the driving force for

Thermodynamics of Solid-State Sintering Process

- Densification requires mass transport, and hence solid-state sintering is carried out at temperatures where material transport due to diffusion is appreciable.
- However, densification will not occur if material transport occurs by surface diffusion.
- In sintering, the driving force is the overall decrease in the surface free energy of the compacts.
- This occurs by replacing the high-energy solid-vapour interfaces with the low energy solid-solid (i.e. particle- particle) interfaces of free energy.
- This surface energy reduction results in the densification of the compact.
- But the reduction in surface energy may also

- Though the overall decrease in free energy is a necessary condition for sintering, the chemical potential differences between the particle and the neck is also a must for sintering to proceed.
- The change in surface free energy during sintering is irreversible.
- Thus sintering can be considered as an irreversible thermodynamic process.
- Therefore higher the driving force for sintering, higher the sintering activity.
- Important factors promoting higher surface area include finer particle sizes and rough surfaces.

Stages in Solid State Sintering

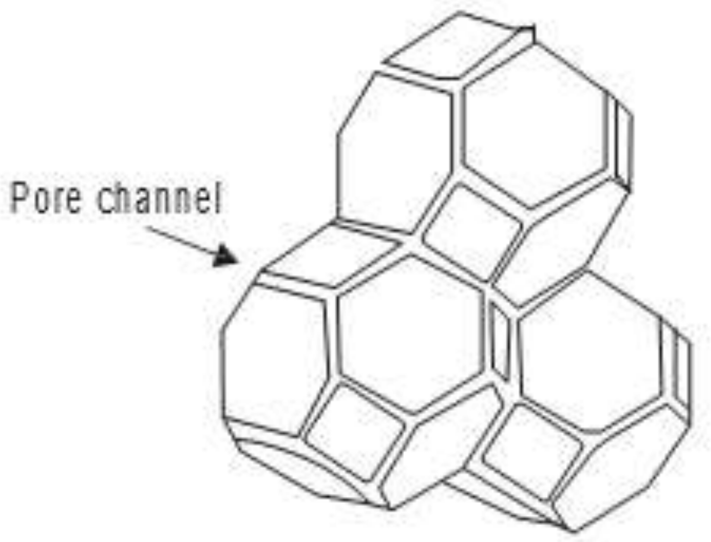


(i) particles in contact

(ii) formation of necks, grain boundaries and pore

(iii) final sintered geometry

- In the **first stage**, necks are formed at the contact points between the particles, which continue to grow.
- During the initial stage, rapid neck formation and neck growth takes place in the powder compact.
- During th re interconnected and the po ular.



Second stage of sintering

- With sufficient neck growth, the pore channels become more cylindrical in nature.
- The interfacial energy is the driving force during this stage.
- The curvature gradients near the necks are responsible for the mass flow during this stage.
- The curvature gradient is high for small neck size leading to faster sintering.
- With sufficient time at the sintering temperature, the pore eventually becomes rounded.
- As the neck grows, the curvature gradient decreases and the sintering rate also decreases.
- During this stage, pore rounding may also occur, without any shrinkage.
- This implies change in pore shape, but no change in pore volume, i.e. pores may become spherical and

- With continued sintering, these cylindrical pore channels become unstable, gradually pinch off and close.
- In this stage, migration of the grain boundaries between the original particles by grain-growth takes place.
- The pores continue to form a more or less connected continuous phase throughout the compact.
- Shrinkage, if any, occurs mainly in this stage.
- In the **final stage**, pore channel closure occurs and the pores become isolated and are no longer interconnected.
- The residual individual pores are located either at the grain boundaries or within the grains.
- In this stage, the porosity does not change and small pores remain even after long sintering times.

Driving Force for Sintering

- The driving force for solid state sintering is the excess surface free energy.
- During sintering, the powder compact tries to reduce the surface energy by transporting material from different areas by various material transport mechanisms so as to eliminate pores.
- Material transport during sintering can occur through gas phase, liquid phase or solid state.
- In solid state sintering, mass transportation can take place either by surface transportation, grain boundaries or bulk transportation.
- Surface transportation can be through diffusion, adhesion or evaporation-condensation or by surface diffusion.
- Viscous flow, plastic flow, grain boundary and volume diffusion are the mechanisms by which bulk

- During sintering, more than one mechanism may operate.
- However, normally one mechanism dominates the others during the different stages of sintering.
- The degree of sintering will depend on the nature of the sintering mechanisms operating, since the mass transported is different for different mechanisms.

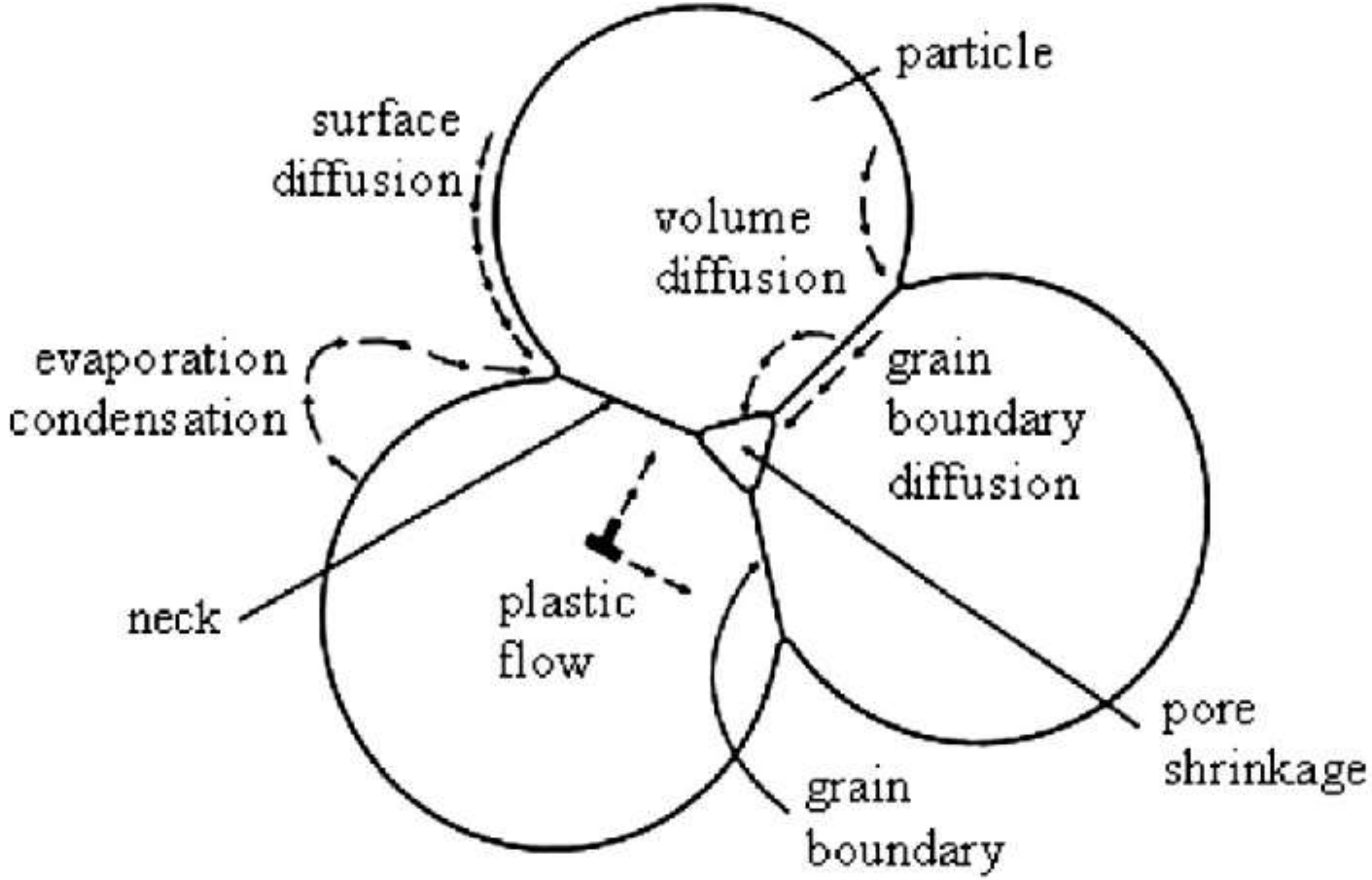
Sintering Mechanisms

(i) Evaporation condensation

(ii) Viscous flow

(iii) Plastic flow

(iv) Diffusion



Evaporation condensation

- This is a simple mechanism, which operates in systems having materials with high-vapour pressure at the sintering temperatures (e.g. chromium).
- For most other metals, this mechanism is not relevant because of the lower values of vapour pressure near their melting point.
- Evaporation-condensation is driven by pressure differences created by surface curvature and changes the shape of the particles and pores.
- The basic principle of this mechanism is that the equilibrium vapour pressure over a concave surface (i.e. neck or a pore) is lower compared to that of a convex surface (i.e. particle surfaces), resulting in mass transport along the vapour pressure gradient, from the convex surfaces to the concave necks.
- Mass transport through the gas phase by evaporation-condensation occurs due to partial pressure difference between a convex surface (larger radius of curvature)

Viscous Flow Mechanism

- Due to the presence of lattice vacancies and is important in the sintering of glasses.
- Based on the surface tension and the glass viscosity and the sintering time.

Plastic flow

- Bulk flow of material by movement of dislocations has been proposed as a possible mechanism for densification during sintering.
- However, any such attempt must include evidence for
- (i) movement of dislocation in the neck region under shear stresses, and
- (ii) generation of new dislocations by surface tension forces.

Diffusion Mechanism

- Diffusion is an important mechanism of mass transport during solid state sintering.
- Diffusion occurs as a result of vacancy concentration gradient.
- This vacancy concentration depends on both temperature as well as the chemical potential gradient.
- In the case of two spheres in contact with one another, a vacancy concentration gradient exists between the two surfaces.
- For the case of a sphere in contact with a flat surface, a vacancy concentration gradient exists between the highly curved surface and

- Of the above mechanisms only viscous flow, plastic flow, volume diffusion and grain boundary diffusion can lead to shrinkage.
- Surface diffusion and evaporation-condensation do not lead to shrinkage but do alter the shape of the neck and particle surfaces by redistribution of atoms as they reach the surface of the neck.
- It has been shown that both the rate of neck growth as well as shrinkage cannot be explained by a single mechanism alone and a combination of two or more mechanisms are needed for neck growth and shrinkage during sintering.

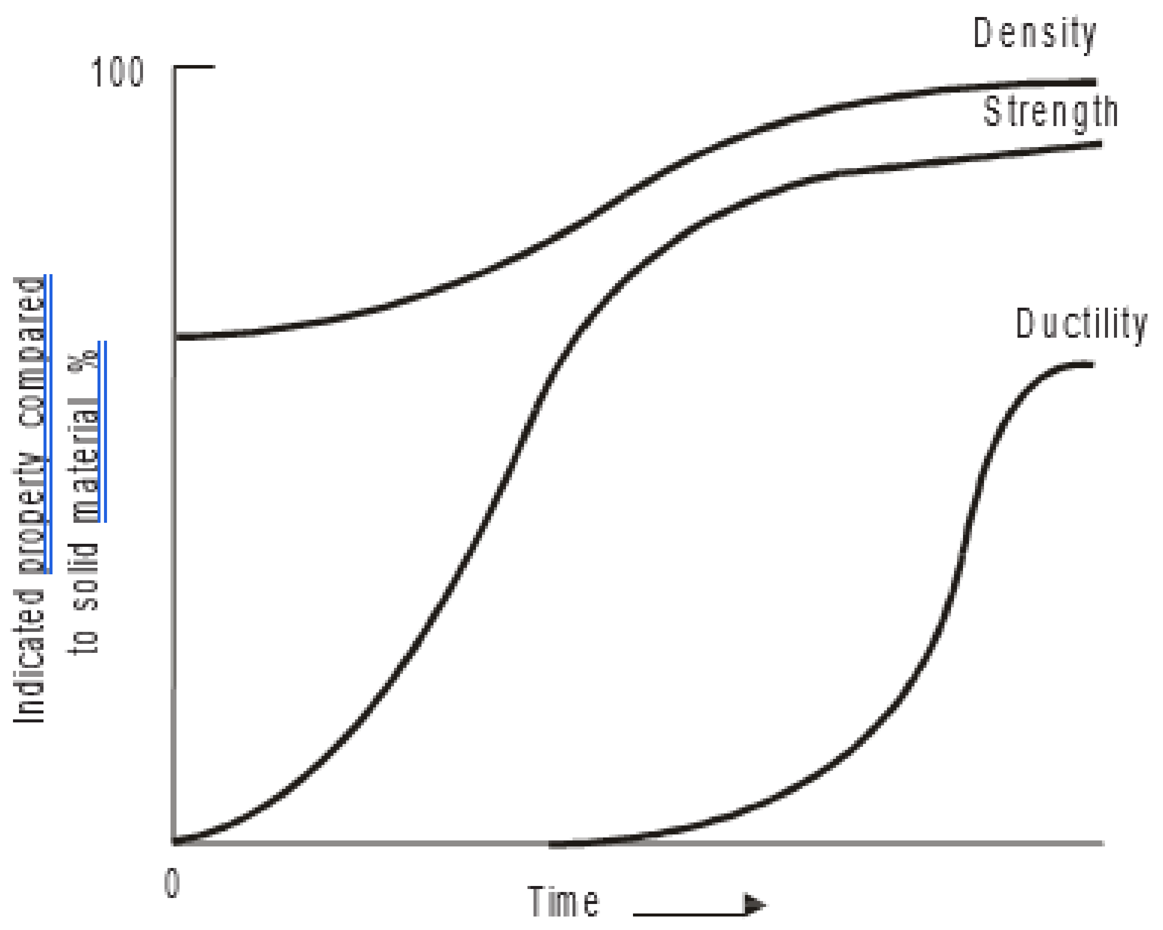
Structure and Property Changes during Sintering

- The main purpose of sintering is the densification of the powder compact.
- Densification is proportional to the shrinkage or the amount of pores removed in the case of single component systems (e.g. metals).
- In the case of multicomponent systems such as premixed powders, during sintering solid solutions may form resulting in an expansion rather than shrinkage
- Hence densification cannot be treated as equal to the amount of porosity removed.
- Expansion of the compact may also take place due to the entrapped gases

- This includes changes in the shape, size and size distribution of grains as well as those of pores present in the initial compact.
- Reduction of surface oxides may occur during sintering in a reducing atmosphere.
- In case of powder mixes containing metals having lower melting point, evaporation or volatilization of these elements can occur and proper care must be taken.
- Other changes taking place during sintering may include recrystallization and grain growth.
- Pore growth also occurs if sintering is carried out for longer times at high temperatures.

- As a consequence, the structural changes result in a change in the mechanical properties such as hardness, strength, fracture toughness, physical properties such as electrical and thermal conductivity, magnetic properties such as the permeability of the material.
- In addition, other changes such as formation of solid solution may lead to a change in the composition.
- The progress of sintering can be monitored by a number of techniques.
- Dilatometry studies can be used to study the densification during the process itself.
- Other methods such as quantitative microscopy or porosimetry studies can be done on samples

Property changes during sintering.



SINTERING OF MULTICOMPONENT SYSTEMS

- Sintering of multicomponent systems in P/M is common, since it is one of the most important methods of forming alloys starting from elemental powders.
- Sintering of elemental powders is advantageous in many ways, for example, it is easy to compact a mixture of iron and carbon powders rather than sintering steel powders.
- Other examples include brass and bronze powders from elemental powders.
- In the sintering of multicomponent systems, the material transport mechanisms involve (in addition to self-diffusion caused by capillary forces), interdiffusion of the components into one another (due to concentration gradient) through vacancy movement.
- Sintering of such systems may also involve the formation of a liquid phase, if the powder aggregate consists of a low-melting component whose melting point is below the sintering temperature (e.g. W-Co, hard

Sintering of Powder Aggregates with No Liquid Phase Formation

- Diffusion occurs and may result in homogenization of composition.
- The extent of homogenization depends on the diffusion coefficients of the elements, which in turn depends on particle size, sintering temperature and time.
- Interdiffusion of constituents will not be equal, with one component diffusing faster than the other.
- For example, in the case of iron-carbon mixture, carbon diffuses much faster into iron than iron into carbon.
- This occurrence of Kirkendall effect during sintering may lead to excess vacancy concentration, which may cause pores to appear and grow, in turn causing the growth of the component.

- In many multicomponent systems, a small amount of second phase, sometimes called as dopants, is added intentionally to increase the densification rates by about 100 times.
- This phenomenon, known as activated sintering, is found to occur in refractory metals like tungsten, molybdenum, tantalum, hafnium and rhenium with nickel or palladium as the activators.
- The significant improvement in sintering characteristics in the presence of dopants has been attributed to the enhanced grain boundary diffusion caused by the presence of dopants at the grain boundaries of these refractory metals.

Liquid Phase Sintering

- Multicomponent sintering in which a liquid phase formed during sintering aids in densification of the compacts.
- Liquid phase sintering employs a small amount of a second constituent having relatively low-melting point.
- During sintering, the temperature is maintained above the melting point of this low-melting constituent, so that it is present either throughout or for a part of the sintering process, as liquid.
- The liquid phase helps to bind the solid particles together and also aids significantly in densification of the green parts.

- Metallic systems include tungsten-copper and copper-tin systems.
- In the case of tungsten-copper, both the elements are mutually insoluble and copper remains as a liquid phase during sintering, while in the case of copper-tin, both are soluble in one another, with tin forming a liquid phase that remains only for a part of the sintering cycle.
- For liquid phase sintering to occur there are three main considerations:

(i) Presence of an appreciable amount of liquid phase.

(ii) Appreciable solubility of solid in

Main stages

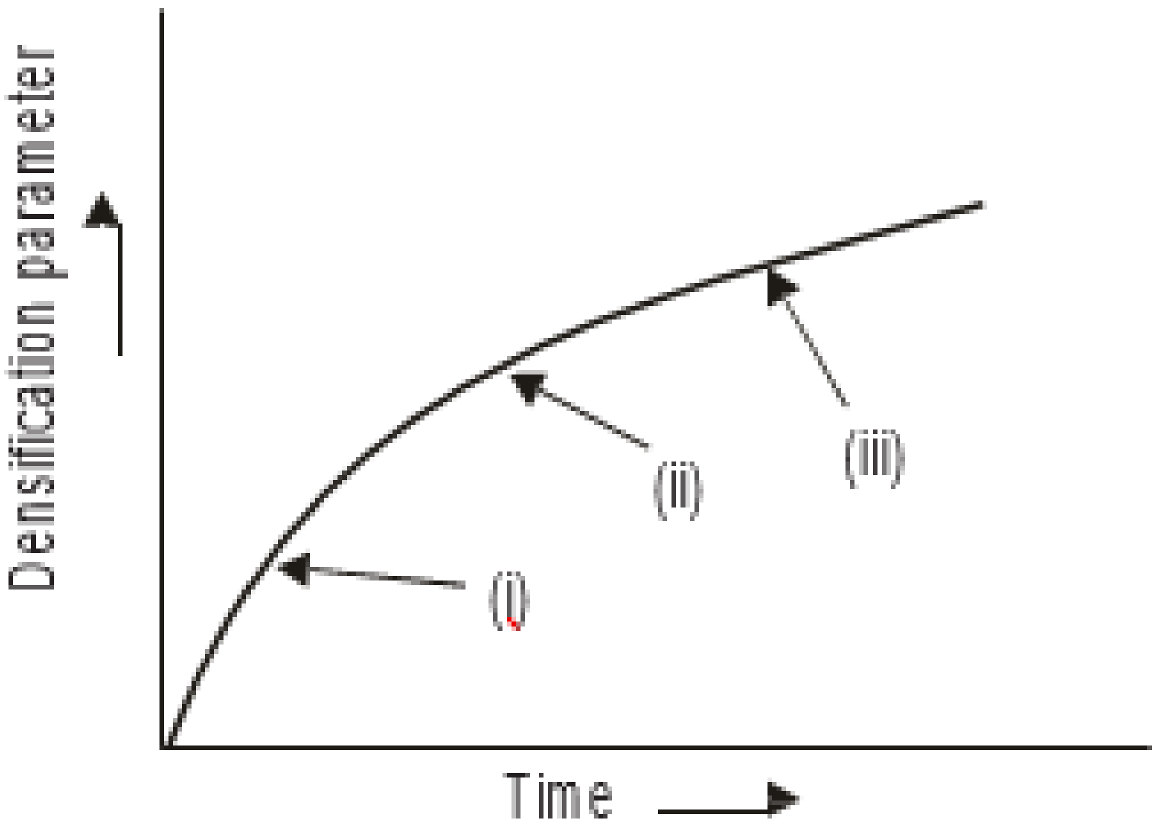
(i) Initially particle rearrangement occurs once the liquid phase has formed.

The solid particles flow under the influence of surface tension forces. This gives rise to efficient packing.

(ii) Solution and reprecipitation process: In this stage, smaller particles dissolve from areas where they are in contact.

This causes the particle centres to approach causing densification.

The dissolved material is carried away from the contact areas and reprecipitate on larger particles.

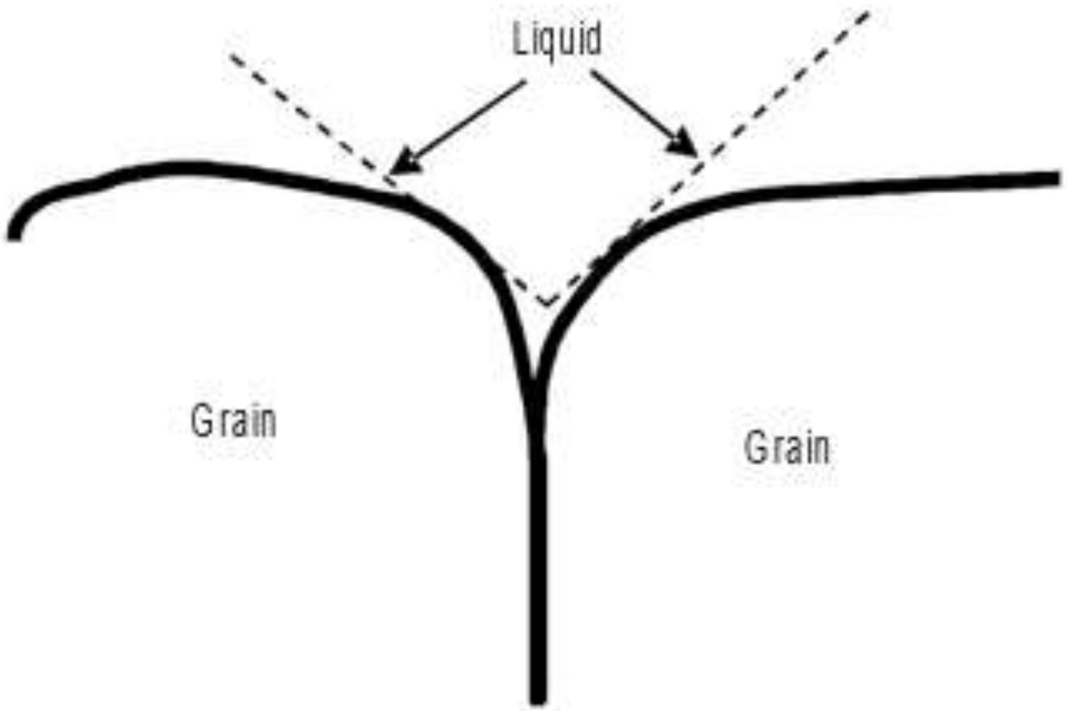

Densification parameter
(i)
(ii)
(iii)
Time

- In another form of liquid phase sintering known as transient liquid phase sintering, the liquid formed is temporary in nature.
- In this case, the liquid continues to take the solid into solution until it reaches a point when the solidus boundary is reached and the liquid resolidifies.
- An example of such a system exhibiting transient liquid phase sintering is 90Cu-10Sn bronze used in preparing self-lubricating bearings.
- During sintering, the tin melts first, forms solid solution with copper and quickly resolidifies.
- Liquid phase sintering will be effective only

Mechanism of Liquid Phase Sintering

- The mechanism of liquid phase sintering is complicated because chemical forces are also acting during sintering.
- A solid skeleton forms similar to solid state sintering during the final stage.
- In the microstructure of the final product, two different phases can be clearly distinguished.
- In this case, the solid particles are coated by the liquid in the initial stage. In liquid phase sintering, the grains (solid) are separated by a liquid film.
- Because of this, the dihedral angle (q) is important

Dihedral angle between two solid particles and liquid phase in contact with them



- Since the driving force for diffusion is the reduction in surface energy, the surface energy term is very important.
- The surface energy for the above system (solid-liquid-vapour) is given by the equation

$$q = g_{s\text{-}s}/2g_{l-s}$$

- where g_{s-s} and g_{l-s} are the interfacial energies between two solid particles and liquid-solid interfaces respectively.
- For complete wetting q should be 0°.

- When the dihedral angle is zero, it means that the two liquid-solid interface can be maintained at a lower energy than a single solid-solid interface.
- A negative pressure gradient is developed because of the small negative radius of curvature of the liquid droplet.
- This forces the particles to come together.
- At the contact point, compressive stresses are developed.
- This compressive stress leads to increased dissolution of a material at the contact points and their movement away from contact points cause the centre of the particles to come together giving rise to densification.
- Shrinkage will occur only if the liquid wets the solid and the solid has a limited solubility in the liquid.

- If the dihedral angle is positive (i.e. q > 0°), then grain boundaries may appear between the particles and an aggregate of two or more grains will be established.
- This situation will lead to the formation of a rigid skeleton, which will hinder further densification.

Conditions for effective liquid phase sintering to be effective

- Initial particle size must be small for solid as well as liquid phase sintering. Small particles have higher surface energy/unit volume and so the driving force for sintering is large.
- Temperature control is necessary, since any diffusion that occurs, is only through liquid phase. Diffusion coefficient in liquid phase does not vary much with temperature. However, too high a temperature may lead to slumping and subsequent deformation of the compact.
- The amount of liquid phase must be kept as

Advantages of Liquid Phase Sintering

- Enhanced kinetics due to the presence of liquid phase, i.e. faster atomic diffusion compared to solid state sintering and hence faster densification.
- The capillary attraction due to wetting allows rapid densification.
- Reduction in interparticle friction leads to faster rearrangement of particles.
- Presence of liquid phase aids in material flow and better packing.
- Sharp edges and corners are dissolved in the liquid resulting in efficient packing.
- Use of additives with lower melting temperatures than the base metal powder enables sintering to take place at lower temperatures.
- Processing of difficult to sinter materials such as

Limitations of Liquid Phase Sintering

- Proper control is necessary. Too much of liquid phase may cause the collapse of object under its own weight.
- Enhanced kinetics can lead to grain growth.
- Possibility of grain boundary segregation can lead to deterioration of creep properties.
- The control of microstructure is more difficult as the liquid phase tends to increase grain growth.
- The high-temperature properties of the sintered material are lowered due to the presence of the liquid phase at the grain boundaries.
- Shrinkage is also more, though may not be present throughout the sintering process.

VARIABLES

- Process Variables

Sintering temperature: Increasing the sintering temperature greatly increases the rate and magnitude of any changes occurring during sintering, due to increased diffusion.

Sintering time: The degree of sintering increases with increasing time, though the effect is small in comparison to the effect of temperature.

The loss of driving force with increasing time at any temperature is one of the reasons why it is so very difficult to remove all porosity by sintering.

It is possible to achieve the desired properties of the sintered parts by shorter sintering times and correspondingly higher temperatures.

However, the maintenance costs and energy consumption of a furnace increase when its operating temperature is raised.

Longer times may also result in grain coarsening, which affects the mechanical properties adversely

Sintering atmosphere: Sintering atmosphere plays a vital role in bonding of the particles and hence in influencing the properties of the sintered part.

Atmospheres not only provide protection against oxidation, but also help to reduce surface oxides, thus, aiding the sintering process.

The proper production, use and control of sintering atmospheres, which are essential for the optimum use of the powder metallurgy process

Material Variables

Particle size

- Generally, finer particle size leads to increased sintering.
- Similar-sized particles have a greater pore/solid interfacial area producing a greater driving force for sintering.
- It promotes all types of diffusion transport, e.g. greater surface area leads to more surface diffusion, small grain size promotes grain boundary diffusion and a larger interparticle contact area to volume diffusion

Particle shape

- The factors that lead to greater and intimate contact between particles and increased internal surface area promote sintering.
- These factors include decreasing sphericity and increasing macro or micro-surface roughness.

Particle structure

- A fine grain structure within the original particles can promote sintering because it favours several material transport mechanisms.

Particle composition

- Alloying additions or impurities within the metal can affect the sintering kinetics.
- The effect can either be deleterious or beneficial depending upon the distribution and reaction of the impurity.
- Surface contamination such as oxidation is usually undesirable.
- Reaction between impurities and either the base metal or alloying additions with impurities at the relatively high temperature may be undesirable.
- Heterogeneous nature of powders (e.g. alloy powders or powder mix) promotes enhanced diffusion and improves sintering kinetics.
- Dispersed phases within the matrix may promote

Green density

- A decrease in green density signifies an increasing amount of internal surface area and consequently a greater driving force for sintering.

- Although the percentage change in density increases with decreasing green density, the absolute value of the sintered density remains highest for the higher green density material.

EFFECTS OF SINTERING

Dimensional Changes

- The fundamental process of sintering leads to a reduction in volume because of pore shrinkage and elimination.
- Following factors are important in this regard.
- **Entrapped gases**: The expansion of gas in closed porosity can give rise to expansion of the compact.

Chemical reactions

- Hydrogen is a common component of sintering atmospheres and can often diffuse through the metal to isolated portions of the compact where it reacts with oxygen to form water vapour.
- The pressure of the water vapour can lead to expansion of the entire mass.
- It is also possible to have reactions that lead to the loss of some element from the sinter mass to the atmosphere such as volatilization, resulting in the shrinkage of the part.

Alloying

- Alloying may take place between two or more elemental powders very often, leading to compact expansion.
- This expansion, which is due to the formation of a solid solution, is often offset by shrinkage of the original porosity.
- Dimensional changes may also occur in a binary system where the rate of diffusion of each metal into the other is different

Shape changes

- Green parts invariably contain variations in green density.
- Such variations can lead to substantial changes in shape because of the strong dependence of sintering, especially shrinkage, on green density.
- Low-green density regions will exhibit a greater amount of shrinkage during sintering.
- For example, a cylinder with a relatively high L/D ratio compacted by a single action method would have a gradually decreasing green density from one end to the other.
- Such as cylinder prepared by a double action method on the other hand, would likely achieve an 'hourglass' shape.
- Many times, the shrinkage (or expansion) is different in two directions, i.e. axial and radial, and must be taken into account during design of the compacting tools.
- Parts with complex and asymmetrical shape, exhibit uneven shrinkage during sintering and in order to achieve the desired tolerance, sizing operation is generally carried out.

Microstructural Changes

- When green compact is sintered, the original particle boundaries can no longer be observed.
- Instead, the structure becomes similar to that of the metal in wrought and annealed conditions, except that it contains pores.
- Typical P/M microstructures are characterized by porosity.
- Pores are of two types, open or closed.
- With progression of sintering, pores continue to shrink.
- At about 5% total porosity, the formation of isolated pores commences.
- In the case of alloys, sintering of elemental powder blends does not always result in homogeneous, well-diffused microstructures.

- Grain growth is another important aspect in sintering.
- Grain growth in a sinter mass can be somewhat different than the conventional metal, since the former can be treated as a two-phase material, consisting of grains and pores.
- Porosity in green compacts as well as in the developing sinter mass represents a very effective hindrance to grain growth.
- The addition of a foreign component in the powder blend may drastically hinder bonding between adjacent particles and the formation of grain boundaries.
- Grain boundary grooves may also tend to inhibit grain growth since the movement of grain boundary away from its groove leads to an

- The control of grain growth so that grain boundaries do not pull away from pores is essential to achieve zero porosity.
- It is important to have as uniformly sized a starting powder as possible so as to reduce separation of pores from grain boundaries.
- Many times a proper additive also plays a vital role.
- A typical example is the sintering of alumina doped with 0.05% MgO, where the additive is preferentially concentrated at the grain boundaries resulting in full densification.

- Many types of phase transformations may occur in the solid state during sintering at a constant temperature or during the cooling of the metal from the sintering temperature.
- Important transformations include martensitic transformation and precipitation hardening.
- Porosity and fine grain structure influence such transformations, an important example being the case of sintered steels.
- Since porosity reduces the thermal conductivity of the mass for a given set of cooling conditions, the prevailing cooling rate throughout a sintered metal is considerably less than for conventional components.
- This is turn would have bearing on martensitic transformation, which is dependent on cooling

- The development of a proper microstructure during liquid phase sintering is very important for attaining the required mechanical properties, in particular ductility.
- During liquid phase sintering, porosity level falls down while grain size increases.
- The shape of the pores varies rapidly during liquid phase sintering.
- During the first stage, the pores are irregular, which later transforms into a cylindrical network of pores before finally attaining a spherical shape.
- The interface energies can change during liquid phase sintering, since they depend on solubility, surface contamination and temperature

POROSITY IN SINTERED P/M PARTS

- Porosity is inherent to most of the P/M parts.
- The green compact prior to sintering can be treated as a two phase material consisting of powder particles and porosity.
- This porosity may or may not be removed during sintering.
- The reduction in free energy may not be too large and generally this results in some residual porosity.
- Thus, the sintered products are frequently more or less porous and this has an influence on the properties of the sintered compact.

- Porosity is a desirable feature in certain P/M applications.
- In fact, some of the applications of P/M are, based on the controlled distribution of porosity, for instance, in oil impregnated bearings and porous filters.
- For different applications, different degree of porosity is desired, for example, about 10 vol % porosity is advantageous in filters.
- By properly selecting the conditions of sintering, the porosity can be tailored to service requirements.
- Pre lubricated bearings are employed in applications where normal lubrication would be either difficult as in the case of sludge pumps or when the lubrication leads to contamination problems in products such as food, pharmaceutical

- However, in most of the other applications, porosity is an undesirable one and has to be eliminated or kept to a minimum.
- For example, in structural ceramics, the residual pores may be the starting points of crack under load.
- Therefore, a reduction of porosity is needed in such applications.
- Typical porosity in conventionally pressed and sintered P/M products varies between 5 and 20%.
- If the porosity levels in the sintered parts are above the specified levels, it can be reduced by secondary operations such as repressing.
- In case very high or close to theoretical densities are needed, as for structural parts, recourse to costlier operations such as sintering

Factors affecting porosity

- Compaction pressure and time.
- Compaction temperature: Die compaction invariably results in significant amount of porosity, whereas hot pressing or HIP result in lower porosity levels due to enhanced material transport due to diffusion as well as plastic flow of material at high temperatures. Hot compacted P/M parts have a higher density and ductility than that of a cold pressed and sintered part.
- Shape of the part.
- Powder particle characteristics such as size and shape.
- Material constitution: Alloys show different sintering characteristics than metals and densification accordingly varies
- Sintering temperature and time: Sufficient temperature and time are needed for good densification.
- Secondary and finishing treatments such as repressing and infiltration.
- The type and amount of lubricants used have a significant influence on the level of porosity in the sintered part. Presence of lubricant affects sintering and the dissociation products from lubricant can lead to porosity.

Need for Sintering Atmospheres

- Most of the metals and alloys react with the gases present in the environment even at room temperature, but more vigorously at higher temperatures.
- The most important reason for using special sintering atmospheres is to provide protection against oxidation and reoxidation of the powders being sintered.
- Green compacts must also be protected from other undesirable reaction like decarburization, as in the case of steel.
- There are many other ways in which a sintering atmosphere can influence the basic sintering process.
- By reducing the surface oxides, the atmosphere may create highly reactive metal atoms.

- Sintering atmosphere, usually a gas or a mixture of gases can also produce undesirable situation, either by reacting with the metal or by getting entrapped within the porosity and preventing full densification.
- Sintering atmosphere plays an important role in determining the final properties of the component.
- The type of material being sintered as well as the properties required influences the type of atmosphere to be used during sintering.

Functions of a Sintering Atmosphere

- Preventing undesirable reactions (oxidation, decarburization or carburization.) during sintering. For example, oxidation of metals occurs if the voids on the surface of the compact are exposed to the atmosphere at the sintering temperature, This in turn hinders the bonding of particles and consequently the sintered properties are poor.
- Facilitating the reduction of surface oxides. For example, use of hydrogen atmosphere during sintering can aid in the reduction of any oxides present on the particle surface, apart from preventing undesirable reactions.
- Facilitating the addition of dopants/other sintering aids/alloying elements which enhance the rate of sintering and promote densification. Such additions are also made to prevent the loss of alloying elements during sintering by maintaining the equilibrium conditions of the atmosphere.
- Aiding the removal of the lubricants or binders.
- Transferring heat to the compacts.
- Controlling the composition (e.g. final carbon content in

- An important requirement of such an atmosphere is that it must not react with the material being sintered. For instance, nitrogen can form nitrides with certain elements like chromium, aluminium or titanium, which is to be avoided.
- During sintering, the composition of a reducing atmosphere itself may change.
- Therefore, the suitability of particular atmosphere to fulfill the desired functions during sintering is to be decided by the thermodynamic principles.
- For example, the reduction of metal oxides by carbon monoxide or hydrogen depends on the equilibrium constant value, which results in a critical partial pressure ratio.

- This value is specific for a given temperature, and the changes in the value of critical partial pressure ratio will change the conditions from reducing to neutral or even oxidising.
- This can be illustrated in the case of steels.
- In the case of plain carbon steels, reduction by carbon monoxide occurs typically by the reaction:
- Fe + 2CO --- Fe(C) + CO_2
- where the term Fe (C) represents the carbon dissolved in iron.
- The equilibrium constant is then given by

$$K = \frac{(p_{CO_2})a_c}{(p^2_{CO_2})(a_{Fe})}.$$

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- By applying the phase rule to this system, the degree of freedom can be determined to be equal to 3.
- These three degrees of freedom are temperature, pressure and the activities of the components.
- Since temperature and pressure have already been fixed, only the partial pressure ratio given by

$$(p_{CO_2})/(p^2_{CO_2})$$

- a_C can be varied to vary the value of K – equilibrium constant.
- In practice, the ratio is varied to give a specific a_C.

- If the compact has more carbon (usually as graphite), then some of the carbon gets removed (i.e. decarburization)
- If the compact has less carbon, then it picks up some carbon from the atmospheres (i.e. carburization)
- If the carbon content of the compact equals the carbon potential of the atmosphere, then the atmosphere will be neutral (i.e. no carburization or decarburization).
- Thus, it is necessary to control the partial pressures to obtain the desired conditions (reducing or neutral).

- Another important factor in the use of sintering atmospheres is the dew point.
- The dew point gives the temperature at which water vapour will condense and is a measure of the moisture content in the sintering atmosphere.
- A dew point of 70°C corresponds to 1 vol% water content which is oxidizing for most metals, while a dew point of -42°C corresponds to 0.01% vol% water vapour which is reducing for most metals.
- Control of dew point is very important during sintering, particularly in steels.
- In the case of steels, sintering using endogas atmosphere needs close control of dew point or carbon dioxide control to achieve good results.
- The relationship between the dew point and the carbon content of the steel has been established and can be used as a guide to control the sintering.
- In general, as sintering temperature increases, the decarburizing effect of water vapour

- The atmospheres used can be grouped under three categories:
    (i) reducing
    (ii) neutral
    (iii) oxidizing atmospheres.

In commercial practice, a wide variety of atmospheres are used, the choice of which is based on the material being sintered as well as the final property requirements of the component.

Nature of Atmosphere -Examples

- Oxidizing – Carbon dioxide, air, steam, oxygen
- Reducing – Hydrogen, carbon monoxide
- Nitriding – Nitrogen or ammonia
- Carburizing – Methane or propane
- Decarburizing – Carbon dioxide or steam
- Inert/neutral – Argon, helium, vacuum

Various atmospheres used in commercial applications

(i) Pure hydrogen.

(ii) Cracked ammonia.

(iii) Reformed hydrocarbon gases.

(iv) Synthetic endogas (mixtures of methanol and nitrogen).

(v) Highly diluted endogas.

(vi) Nitrogen-based mixtures (N_2 + H_2) without carburizing addition.

(vii) Nitrogen-based mixtures (N_2 + H_2) with carburizing addition.

(viii) Inert gases (argon, helium), and

(ix) Vacuum.

- Use of hydrogen atmosphere is limited (for sintering of carbides and AlNiCo magnets) because of cost and handling problems.
- Dissociated ammonia is a low cost alternative for the hydrogen atmosphere and can be readily synthesized from oxygen or water vapour.
- However, this atmosphere should not be employed where nitrogen is to be avoided.
- Use of combusted hydrocarbons can also be used as a reducing atmosphere.
- Exogas and endogas are among the most commonly used sintering atmospheres owing to their lower cost.
- They are produced by partial combustion of hydrocarbons by a careful control of the air-to-gas ratio and a wide range of compositions can be readily obtained.
- Such atmospheres may require special equipment like generators and may add to the initial cost

- Apart from these atmospheres, a number of nitrogen-based atmospheres, containing predominantly nitrogen along with hydrogen are also employed.
- They provide a clean atmosphere, very low water vapour content and little variation in composition but are more expensive than exogas or endogas.
- It is possible to have a combination such as hydrogen and methane that is carburizing and reducing at the same time.
- Inert atmospheres like argon and helium are very expensive and are employed only in specific cases like sintering of reactive

- Vacuum sintering also requires expensive equipment and used only in special cases like tungsten carbides, stainless steels, titanium, tantalum, vanadium, zirconium, thorium and cermets because of the tendency of the constituents to form carbides, nitrides or other compounds.
- In the case of stainless steels, chromium oxide must be removed.
- This requires the use of hydrogen to convert oxygen into water vapour, which is removed from the furnace.
- Steels containing aluminium, titanium, chromium and tantalum can form nitride, which degrades the properties when sintered in a nitrogen atmosphere.
- Sintering requires effective control over heating rate, sintering time and temperature apart

Hydrogen

- Reducing atmospheres are, by far, most commonly used for sintering metal parts.
- Pure hydrogen is an excellent reducing gas
- Hydrogen atmosphere is very expensive and is also highly inflammable.
- Hydrogen has a high rate of flame propagation and burns immediately on contact with air.
- It is lighter than air (specific gravity 0.069) and gets easily displaced.
- The thermal conductivity of hydrogen is also very high, resulting in acceleration of both the heating and cooling rates of parts in furnaces.
- This leads to relatively large heat losses in furnaces using hydrogen atmosphere

- Its reactivity with oxygen helps in the sintering of alloys containing elements that form stable oxides (e.g. chromium in steel or stainless steel).
- To reduce the oxide, the hydrogen reacts with the oxide to form water vapour, which is swept away in the furnace.
- If decarburizing conditions are required during sintering, then hydrogen gas must be free from hydrocarbon, carbon monoxide and carbon dioxide gases.
- Hydrogen gas in the presence of carbon can form hydrocarbons, which are ideal for sintering of parts requiring a carburizing atmosphere.
- If carburizing conditions are required while using hydrogen, graphite parts such as graphite

- Hydrogen gas can be produced either by (i) reacting water and methane or other hydrocarbon gases or by (ii) electrolysis of aqueous salt solutions.
- When hydrocarbon gas is reacted with water or steam, the resulting product is a mixture of hydrogen and carbon monoxide.
- The latter is converted to carbon dioxide and absorbed using chemicals to give nearly pure hydrogen.
- Commercial grades of hydrogen gases are always associated with small amounts of oxygen and moisture, which may interfere with sintering by forming adherent oxide films.
- Oxygen is removed by passing the gas over a heated catalyst like platinum or over a heated bed of copper filings

- Removal of water vapour present in hydrogen is done using activated alumina or zeolite.
- The moisture can also be removed by refrigeration which is relatively more expensive, especially if the dew point required is less than –10°C.
- Traces of moisture in purified hydrogen may also be removed by the use of getters (substances having high affinity for oxygen) in the furnaces where hydrogen is used.
- The water vapour content in any gas is usually specified in terms of 'Dew Point.'
- Dew point is defined as the temperature at which the water vapour in the gas begins to condense on cooling.

- When the gas is cooled to low temperature, the amount of water vapour in the gas decreases steadily (i.e. the gas becomes increasingly 'dry').
- For very low water vapour content, of the order of 0.02%, the dew point of the gas is –40°C.
- Commercial sintering atmospheres can be dried upto –40°C through the use of activated alumina.
- Further reduction in dew point requires the use of zeolite.
- Hydrogen atmosphere is used for the reduction of oxides of iron, molybdenum, tungsten, cobalt, brass, bronze, nickel, stainless steel and tungsten as well as for annealing of electrolytic and carbonyl iron powders and for

Variation of Water Vapour Content As a Function of Dew Point of Gases

Water vapour (Vol %)	7	35	1.75	0.82	0.60	0.37	0.15	0.06	0.02
Dew point (◦C)	37	27	15	5	0	−7	−18	−29	−40

Reformed Hydrocarbon Gases

- These are cheap and widely used sintering atmospheres obtaining by controlled combustion of hydrocarbon fuels like methane, propane and butane.
- Hydrocarbon gases are generally classified into

(i) exothermic gas or exogas,

(ii) endothermic gas or endogas.

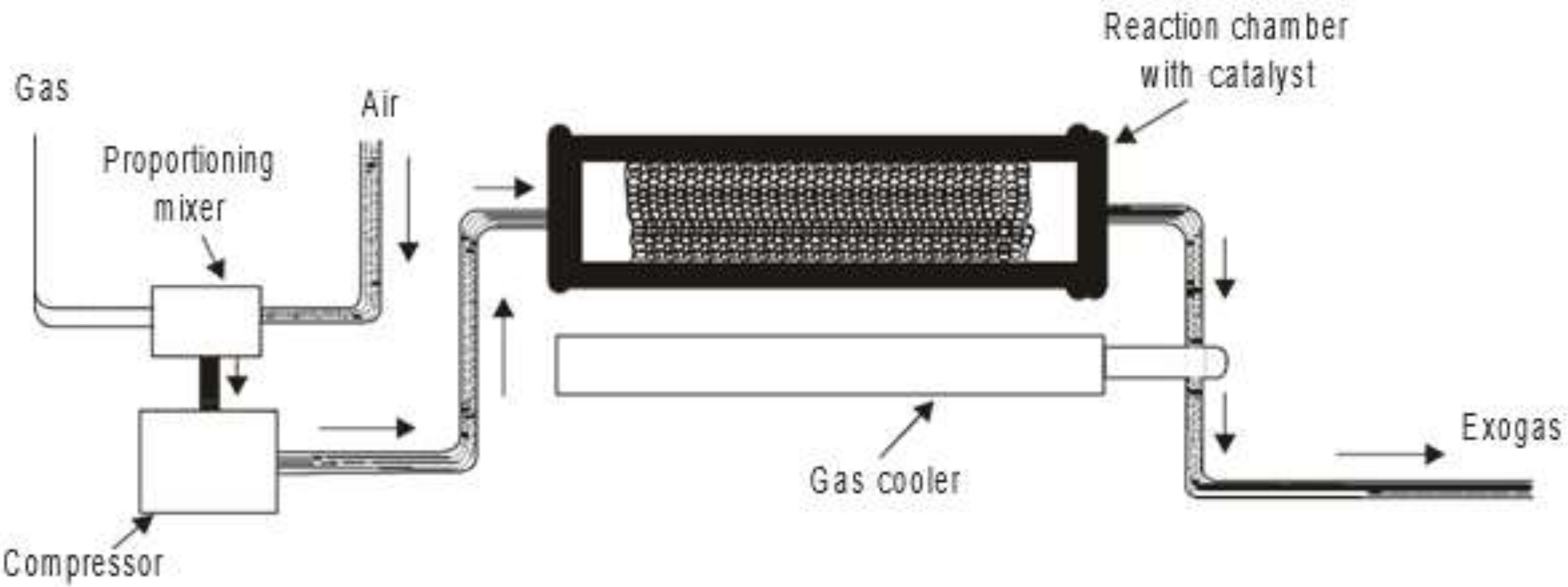
Exothermic Gas

- When a hydrocarbon gas is burnt with sufficient air for complete combustion, the resulting gas contains mostly nitrogen (70–90%) with about 5% or less of hydrogen.
- This reaction being exothermic in nature, the atmosphere is called an exogas and is one of the cheapest atmospheres available.
- However, owing to its lower reducing potential, removal of oxides from the powder compacts is limited leading to lower sintered strengths.
- It is to be remembered that since a large number of components are iron or steel based, the carbon potential of the atmosphere should be in equilibrium with that of the steel and must be carefully controlled to avoid undesirable

- Removal of carbon dioxide and water vapour from exogas results in purified exogas.
- Exothermic gases can be produced from petrol, crude oil as well as liquefied petroleum gas.
- From crude oil, an exogas containing 20-30% CO and H_2 can be obtained.
- Exothermic gas is produced in specially designed generators, where fuel gas and air are mixed in such a ratio that incomplete combustion takes place.
- The heat generated in the combustion chamber however, is sufficient to support reaction.
- At an air to fuel ratio of 6:1, combustion is practically complete.

- The resulting mixture is relatively inert to metals such as hot copper, tin, silver.
- It will, however, oxidize hot iron and reactive metals because of the high proportion of carbon dioxide and water vapour as opposed to extremely low percentages of reducing component hydrogen and carbon monoxide.
- This gas is known as lean exothermic gas and has very limited application in Powder Metallurgy.

Schematic diagram of exothermic gas generator

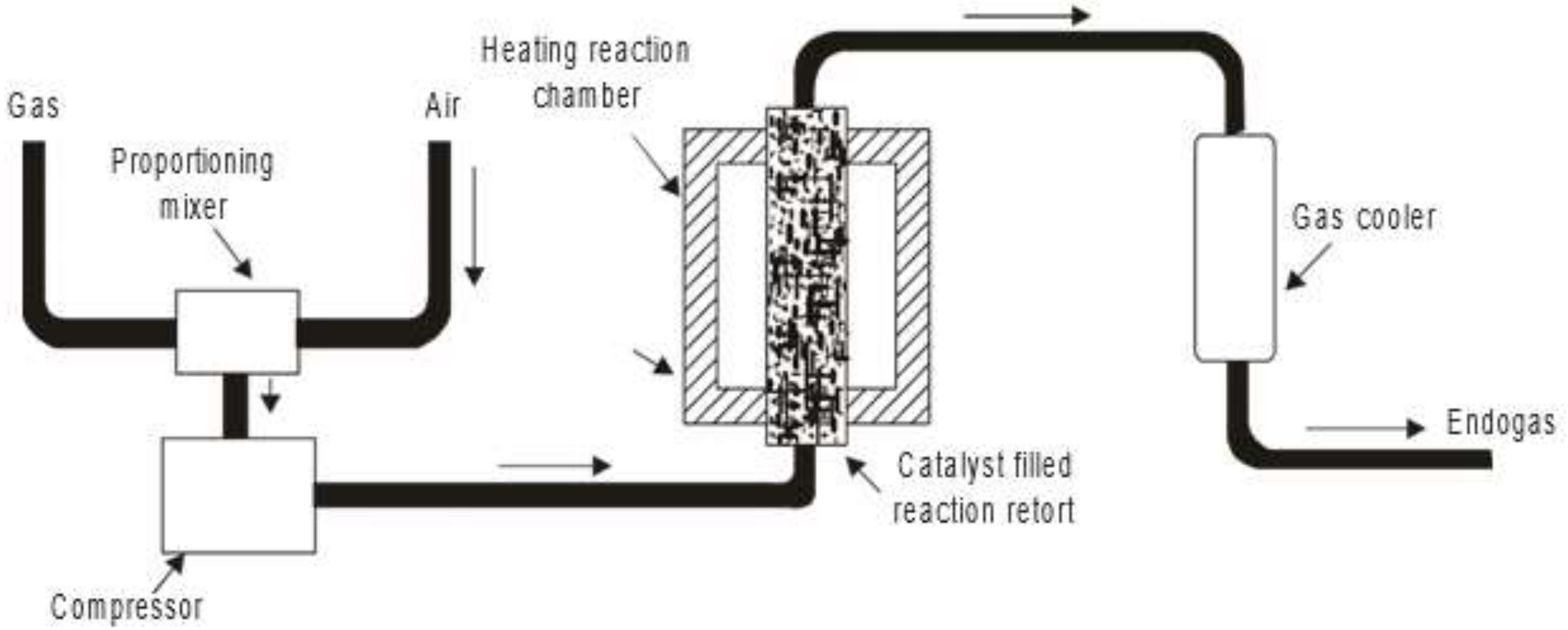


Endothermic Gas

- Endothermic atmospheres are low-cost atmospheres formed via a catalytic reaction of natural gas (about 93% methane, a reducing gas) and air with low air to gas ratio.
- Hydrocarbon gas and air in various ratios are partially combusted in a gas generator operating at 1,000°C to 1100°C.
- An endogas is so called because of the endothermic nature of the reaction and hence the need to supply external heat.
- This can be produced when hydrocarbon gas such as methane, butane or propane is burnt with air to give a gas containing about 45% hydrogen along with nitrogen, carbon monoxide, carbon dioxide, water vapour and small amounts of uncombusted hydrocarbon gas.
- It is richer than exogas in CO, H_2 and CH_4.
- This gas is not only strongly reducing but is also carburizing.
- Endothermic gas is most suitable for sintering compacts of iron-carbon and other alloy steels.
- The composition of the endogas is controlled so that it is in equilibrium with the carbon potential of the steel to be sintered.

- Endothermic gas formation is facilitated over a clean catalyst such as nickel oxide in an externally heated chamber.
- The water vapour present in the gas must be dried to a dew point of less than 0°C for satisfactory sintering of ferrous parts.
- Most of the water vapour produced is removed by cooling the gas so as to reduce the water vapour content from about 5 to 15% to below 1%.
- Water content in the gas mixture is very important for sintering.
- Higher water content of 5 to 15% makes the atmosphere oxidizing while an atmosphere containing 0.1% water vapour with a dew point of -40°C is a strong reducing atmosphere

Schematic diagram of endothermic gas generator



- Typically, the generator is operated at temperature of about 1,060–1,100°C.
- The factors that influence the composition of the produced endogas are:

(i) Temperature in the cracking zone

(ii) The air-to-gas ratio

(iii) The efficiency of the catalyst

(iv) The time in the cracking unit

- The temperature of the cracker should be kept close to the maximum temperature in the sintering furnace.
- Otherwise, the gas may not be stable in the sintering furnace causing soot formation or result in uneven properties of the sintered steel.
- At lower temperatures, CO is strongly carburizing and reducing.
- At higher temperatures, however, the action of gas is weaker.
- The following reactions take place:

(i) Producer gas reaction:

- $CO_2 + C \rightarrow 2CO$

(ii) Carburization reaction (carbon monoxide with iron):

- At lower temperatures, the producer gas reaction is generally the most prevalent and result in the soot deposition on the sintered parts which can be suppressed by fast heating and cooling.
- However, at higher temperatures, say at 800°C, the carburizing action is dominant.
- The effect of methane is different from that of carbon monoxide.
- Methane, with increasing temperature, increases the reducing action and the rate of carbon deposition.
- The rate of decarburization and recarburization determines the final carbon content at or near

- The dew point or the carbon dioxide content in the endogas can control carbon content.
- Most sintering endothermic atmosphere is done at dew point ranging from –5 to 15°C.
- Endogas is reducing to iron above 260°C, because it typically contains 40% hydrogen and 0.8% water vapour (dew point 5°) and gives a hydrogen to water ratio of 50 to 1.
- Although this ratio gives reasonable oxide reduction, other atmospheres notably those based on hydrogen or dissociated ammonia or synthetic nitrogen based atmospheres can provide higher ratios.

- Recently, a mixture of methanol (or methyl alcohol) and nitrogen has been used as controlled atmosphere.
- When this mixture is introduced into the sintering zone of the furnace, at high temperature the following reaction takes place:
- $CH_3OH \rightarrow CO + 2H_2$
- As the ratio between CO and H_2 is the same as that of endogas from methane, by mixing nitrogen, it is possible to form an endogas inside the furnace itself, thus eliminating the use of a separate generator.
- By adding higher amounts of nitrogen, a highly-diluted endogas can also be obtained.

Nitrogen Based Mixtures

- Nitrogen and nitrogen based atmospheres are widely used in sintering of metals and alloys.
- Nitrogen is inert to most common metals and alloys.
- It is non-flammable, it is also used as a safety purge for flammable atmospheres.
- The main constituent of the nitrogen based system is molecular nitrogen.
- Nitrogen is most commonly produced through air separation, i.e. liquefaction and fractional distillation.
- Nitrogen provides a cheap and useful atmosphere

Characteristics of Nitrogen

- It is very dry, having a dew point of less than −65°C.
- It is very pure, having less than 10 ppm of oxygen.
- It is essentially inert to materials most commonly sintered and to furnace components such as muffles, conveyer belt, heating element, radiant tube and fixtures.

- In actual practice, furnaces are not very tight and some air does get in.
- Nitrogen by itself does not control the resulting surface oxidizing and decarburizing.
- **Chemically active gases are**, therefore, added to the nitrogen, when the atmosphere has to perform functions requiring transfer of some element such as carbon from the atmosphere to the component being treated or such as oxygen from the oxide to the atmosphere.

Active ingredients

Oxide reducing agents: The most desirable ingredient for reducing surface oxides is hydrogen.

It can be derived from liquid hydrogen storage tanks or dissociated ammonia.

It can also be derived from endogas or dissociated alcohols.

Methanol dissociates or cracks at temperatures above 815°C to produce H_2 and CO in the same ratio (2:1) as normally found in the endo gas generated from natural gas:

- $CH_3OH \rightarrow CO + 2H_2$ + small amount of CO_2, H_2O and CH_4

- Dissociated or cracked methanol when mixed with appropriate amounts of nitrogen can produce an atmosphere almost identical to endogas.

Carburizing agents: The most desirable ingredient to effect carburization is CO.

It can be derived from endogas or dissociated alcohols such as methanol or by reducing a hydrocarbon such as natural gas or propane and an oxidant such as water in the carburizing zone.

It is generally found that in order to maintain effective carburizing rates, the CO level in the carburizing zone should not be less than 10 percent.

Oxidants: Small but controlled amounts of oxidants such as CO_2, H_2O, O_2 or some combination of these gases can be added to nitrogen at selected sections of the furnace to provide decarburizing, oxidizing

- Manufacturing of nitrogen consumes less energy than producing an equivalent volume of endogas: 44% less energy in the liquid storage process and 80% less energy in the on-site separator process.
- Nitrogen allows smaller volume of atmosphere to be used because it permits reduced flow rates, increase in production rates, lower part rejection and increased furnace utilization.
- With the proper choice of enrichment gas, nitrogen-based atmospheres can be used to sinter and infiltrate iron, carbon steel and other ferrous and nonferrous alloys.
- Stainless steel and refractory metals can

- Compositions containing 80 to 97% nitrogen are common in use.
- The higher the nitrogen content the lower is the cost and flammability.
- Nitrogen-hydrogen mixture is non-flammable below 5% hydrogen. .
- Use of nitrogen is avoided if the steel contains titanium, aluminium, chromium or tantalum as it can form nitrides.

Dissociated Ammonia

- Dissociated ammonia is another common atmosphere consisting of hydrogen and nitrogen.
- Dissociation is carried by the evaporation of liquid ammonia in a vapourizer at high temperature (900–1,000°C) in the presence of a suitable catalyst.
- Ammonia gas for dissociation is obtained from ammonia cracking units and the gas has a dew point between –60°C and –70°C and delivered as a liquid under pressure.
- Liquid ammonia from a storage tank enters a vapourizer at high pressure where the heat

Ammonia gas generation unit

- The pressure of the vapour is then reduced in an expansion valve and the low-pressure vapour passes through a dissociator element filled with catalyst to give nearly a pure atmosphere of hydrogen and nitrogen.
- Traces of moisture (< 0.1%) can be removed passing it over activated alumina.
- $2NH_3(g) \rightarrow 3\ H_2(g) + N_2(g)$
- Cracked or dissociated ammonia contains 75 vol % hydrogen and 25 vol % nitrogen.
- It has high purity and is sufficiently dry for use in furnace directly.
- The dissociated gas is highly flammable at elevated temperatures.
- Its specific gravity is 0.295 and thermal

- Typical applications of dissociated ammonia include sintering of many steels, brass, copper, iron-copper, tungsten and tungsten alloys, aluminium and its alloys tungsten and stainless steels.
- This atmosphere has several advantages over hydrogen atmosphere and can be substituted for hydrogen in many cases.
- Main advantages include lower cost, high purity and simple operation procedures.
- The dissociated gas containing hydrogen and nitrogen is inflammable and has high thermal conductivity.

Inert Gases

- Inert gases such as argon and helium are used as sintering atmospheres in very special cases as in the case of the sintering of extremely reactive metals.
- The limited use of inert gas is due to their relatively high cost and their lack of reducing ability.
- Inert gases, mainly argon and helium are used for sintering of reactive and refractory metals like superalloys, titanium and zirconiums as well as a back fill in vacuum furnaces.
- Argon and helium are produced cryogenically from air.

Vacuum

- Vacuum sintering is widely used in sintering and involves evacuation of the furnace chamber to the desired level of pressure and maintaining it throughout the operation.
- Vacuum retains the proper chemistry of the parts during sintering.
- It is often more economical than atmosphere gases, particularly bottled gases like Ar and He.
- The only operating costs involved in producing the vacuum are for electrical energy and oil for the pumps.
- Vacuum pumps commonly use mechanical pumps and oil vapour pumps.
- Vacua are generally classified with four ranges: rough (133 Pa), medium (133- 0.133 Pa), high (0.133 to 1.33 x 10^{-5} Pa) and ultra-high vacuum (below 1.33 x 10^{-5} Pa).
- For pressures up to medium level, rotary pumps can be used.
- High and ultra high vacuum requires combination of rotary and diffusion pumps.

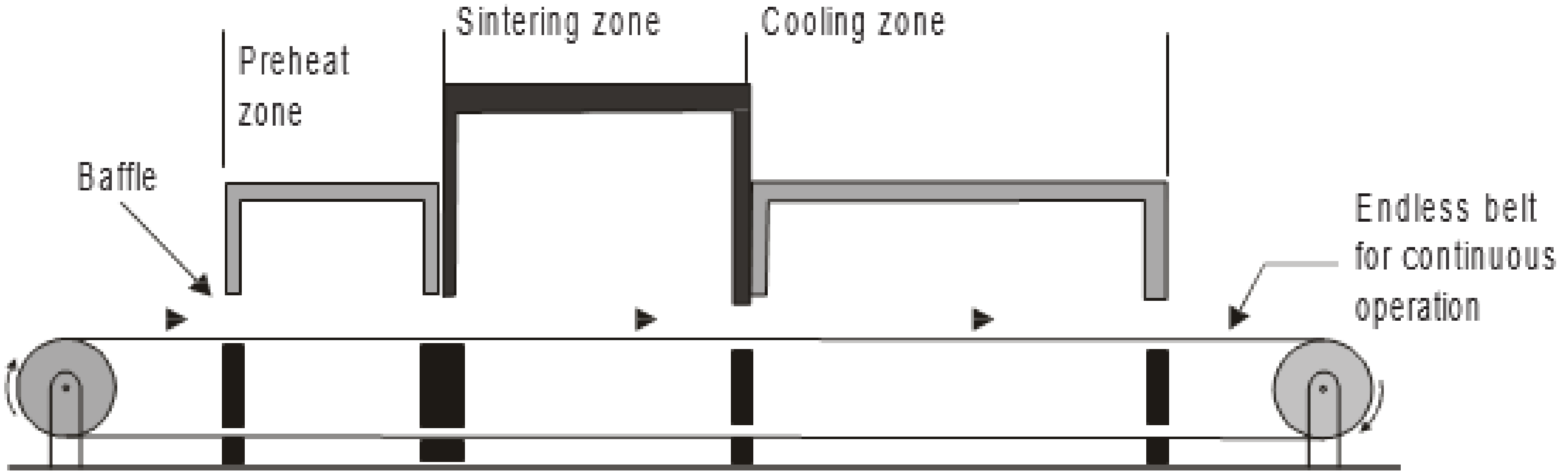
- Vacuum sintering is used for high reactive metals like titanium, tantalum, thorium, vanadium, zirconium, beryllium and refractory metal carbides since these materials react with other gaseous atmospheres forming oxides, nitrides and hydrides.
- It is also used for sintering of stainless steels and tool steels, which require high temperature sintering.
- The requirements of vacuum systems, controlling systems and leak-proof design make vacuum sintering a very costlier choice.
- However, it provides a clean and nonreactive atmosphere giving good, reproducible sintering atmosphere.
- It is also possible to have additions of reactive metals (like titanium, niobium and

- All metallic oxides have a so called dissociation pressure, which is equal to the partial pressure of the oxygen present in the gas atmosphere in equilibrium with the oxide.
- If the partial pressure of the oxygen is lower than this, the compound will be transferred into a lower value oxide or metal and oxygen.
- If the partial pressure of the oxygen is higher than the dissociation pressure, the metal will oxidize.
- Vacuum level and temperature should be selected accordingly.
- During the sintering of an alloy, due to the difference in vapour pressures of the individual metals, selective evaporation of some alloying elements may occur.

SINTERING PRACTICE

- The sintering of P/M components requires a sintering furnace, close control and regulation of its temperature, heating time, heating rate and sintering atmosphere.
- For obtaining consistent and satisfactory sintered components, it is essential to make use of furnaces operating at the lowest temperatures for the shortest times using the cheapest possible sintering atmosphere with lowest maintenance and power consumption as well as the least labour costs.
- A sintering furnace is selected depending upon these factors: the working temperature range, type and purity of the sintering atmosphere during sintering, desired production rate, operation and maintenance costs for a given production.
- The essential parts of a sintering furnace include: (i) a burn-off zone for the removal of lubricants, (ii) a controlled high-temperature sintering zone, and

Continuous belt sintering furnace

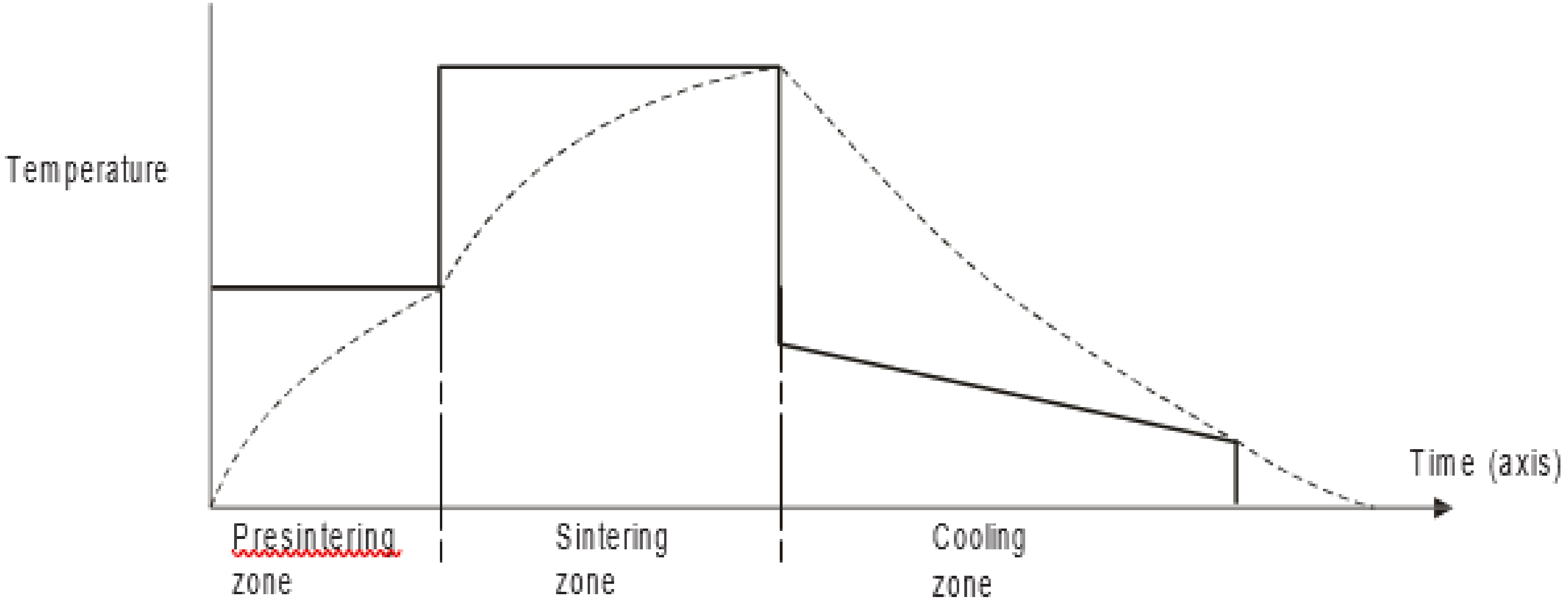


- The first zone, also known as the burn-off or preheat zone, must be capable of working up to a temperature of 400-500°C.
- The green compacts on entering this zone, are heated slowly so as to avoid rapid heating which may result in the expansion of entrapped air and lubricants.
- When the compacts leave this zone and enter the sintering zone, the lubricant must have been completely removed.
- Burn-off chamber is usually separated from the high-temperature sintering zone, with an air gap employing a flame curtain.
- Atmospheres aid in the removal of lubricants

- Decomposition starts at 150°C and is completed at 500°C.
- Polymers used as lubricants decompose into methane, ethane, butane, propane, CO, CO_2, H_2O during heating.
- When burnout is incomplete, sooting may occur along with discolouration of the components.
- Sooting occurs because of rapid rate of heating without sufficient oxygen or moisture.
- During heating, the polymer directly forms carbon, which deposits on the compacts.
- This in furnaces with high-heating rate, low-atmosphere flow rates and for large or high-density parts.
- For complete removal of polymer, slow heating over the range of temperatures where burn out take place is recommended.
- However, it must be ensured that the compacts are at 650°C before entry into the high temperature portion of sintering furnace.

- The second or the high-temperature sintering zone must be longer in length in relation to the burn off zone to provide sufficient time to obtain the desired density and strength in the sintered parts.
- The actual sintering time will vary depending on the starting material, its purity, the sintering temperature and the atmosphere used.
- It is advantageous to obtain the desired properties by shorter sintering times using higher temperatures.
- The cooling zone of a sintering furnace is made up of two sections.
- The first section is a short insulated cooling zone, which allows the sintered parts to be cooled slowly in order to avoid thermal stresses.
- A relatively longer water-jacketed cooling zone to cool the sintered parts without exposure to air to prevent oxidation follows this insulated zone.

Progress of sintering



- Continuous mesh belt furnaces are commonly used when atmosphere control is needed.
- Large production units use continuous type furnaces, such furnaces employ mesh belt, pusher or roller hearth to move the compact through the furnace.
- Mesh belt type furnaces require large volumes of protective gases as the doors are kept open during operation.
- In the case of roller hearth, the doors are kept closed during operation and consequently heat losses are minimized.
- In addition, relatively lower volumes of protective atmosphere are sufficient.
- Electrical furnaces of the mesh belt type are normally employed for sintering temperatures up to 1,100°C.
- For sintering temperatures up to 1,350°C, silicon carbide heating elements are employed.
- Molybdenum heating elements (requiring hydrogen atmosphere to prevent oxidation) can be used for still higher temperatures.

Sintering Atmosphere Analysis and Control

- Sintering atmosphere must be carefully controlled to

(i) maintain the correct chemistry of the metals or alloys being sintered

(ii) ensure good bonding between the particles, achieve the required density and hence properties

(iii) prevent undesirable reactions which may affect the dimensions of the sintered part (e.g. carburization and decarburization of steels parts).

It is therefore essential to analyze the gas content, specific gravity of the gases, moisture content and carbon potential of the sintering atmosphere as well as to control them.

Infrared analyzers are used to analyze gases such as CO, CO2 and methane.

By measuring the specific gravity of effluent gas from the furnace continuously, the completeness of purge can be identified.

By checking the dew point, the moisture content of the

Gas Analysis

- A complete analysis of the gas can be done using the Orsat type analyzer or infrared analyzer.
- In the Orsat type analyzer, a complete analysis of the gaseous mixture is divided into two phases, namely, the absorption phase and the explosion or burning phase.
- The constituents of the atmosphere such as carbon dioxide, carbon monoxide and oxygen are determined by chemical absorption.
- Portable units available lend themselves to rapid spot analysis.
- Infrared analyzers work on the principle that each constituent (methane etc.) absorbs infrared energy at a characteristic wavelength.
- Any change in the concentration of the particular constituent results in a change in the total energy of a beam passed through the gas.
- This apparatus permits rapid determination of carbon monoxide, carbon dioxide and methane, but oxygen, nitrogen and hydrogen cannot be analyzed.

- Such analyzers are highly sensitive and are successfully

Moisture Analysis

- Dew point analysis is an effective method of determining the moisture content of a gas.
- This is done using a "Dew Cup."
- The apparatus, which consists of a glass thermometer, is kept directly in the path of the gas stream.
- The dew cup is first filled with acetone and the atmosphere gas is passed against the dew cup.
- After about 5 minutes, dry ice is added to the cup with stirring.
- Moisture or dew appears on the surface of the cup and the dew point is read from the


010
DEW CUP

- In a more sophisticated type of indicator, the sample of the gas is compressed, and then quickly expanded.
- In case the gas got cooled down below its dew point due to its rapid expansion, a fog is noticed in the expansion chamber.
- The pressure ratio of the gas is converted to dew point.
- The dew point measurement of a gas either entering or leaving a furnace gives an excellent indication of the carbon potential.
- Presence of excess moisture can lead to decarburizing of the parts in a non-decarburizing atmosphere.
- The measurement is also an indicator whether water-gas reaction is occurring within the furnace atmosphere through the reaction of carbon dioxide

Specific Gravity Analysis

- The specific gravity of gases can be measured and compared against that of air.
- This analysis is especially sensitive to changes in carbon dioxide, since carbon dioxide is much heavier than the other sintering atmosphere constituents.
- In case carbon dioxide is present in the incoming or outgoing furnace atmosphere, specific gravity analyzer could be used for taking corrective measures.
- This analysis is used, for example, in checking sintering furnace atmospheres of purified rich exothermic or rich endothermic gas which are supposed to be free from carbon dioxide.
- By continuously measuring the specific gravity of the effluent gas from the furnace, the extent of purging (flushing) in the furnace chamber can be established.

Carbon Potential Control

- Carbon potential control of sintering atmosphere is important, particularly during sintering of steels.
- Equilibrium between the atmosphere and the carbon potential of the steel concerned will prevent the steel from getting carburized or decarburized.
- As mentioned earlier, both dew point as well as the carbon dioxide analysis of the atmosphere can be used as a measure of its carbon potential.
- The gas sample taken for measurement of the carbon potential should be from the high heat zone of the sintering furnace.

- As the furnace is usually not under positive

- Dew point analyzers are not as sensitive, and are slower in response than infrared analyzers.
- Dew point analyzer is also not suitable for use in multipoint control because of difficulty in purging the water molecules from the instrument in a short time.
- The quick response infrared analyzers are ideal for multipoint control systems.
- Process control is done by referring the equilibrium data of either dew point vs carbon content or carbon dioxide vs carbon content at the sintering temperature.
- These data can be determined by actual measurement and are fairly accurate.
- However, they should be used only as a guide and the final adjustment should be made after the carbon analysis of the steel concerned.

- Each zone of a sintering furnace requires a unique combination of temperature, time and atmosphere composition.
- To achieve this, many times the atmosphere is either diluted or enriched by hydrocarbon.
- To achieve specific carbon content in the sintered steel, it is therefore important to understand the effect of temperature in each zone on the final carbon equilibrium of the concerned steel.

Temperature Control

- Temperature is an important parameter in sintering and must be closely controlled in order to achieve the desired density and properties.
- Too high or too low a temperature during sintering will have adverse effect on the structure as well as properties.
- Temperatures are measured using thermocouples in case of low temperature furnaces and by radiation pyrometers in the case of high temperature furnaces.
- The temperature is then closely held with

Temperature controls used in sintering furnaces

- On-off controllers: This is one of the simplest types of control and is used in both gas-fired and electrically-fired furnaces.
- In the case of gas-fired furnaces, it functions by closing a valve when the temperature goes above the cut-off limit and opens the valve if the temperature falls below the specified limit.
- In the case of electrically-fired furnaces, the controller breaks a contact or makes a contact when the measured temperatures are above and below the cut-off values respectively.

Proportional controllers

- These are more sophisticated instruments used widely in electrically-heated furnaces.
- A proportional controller is a type of feedback controller that adjusts a system's output in proportion to the error between the desired setpoint and the measured process variable.
- It works by continuously calculating the difference (error) between the desired state and the current state, and then adjusting the output signal to minimize this error.

Transformer controls

- This type of control uses a transformer whose AC windings are connected to the mains while a DC winding present in the transformer is connected to the thermocouple.
- Any changes in temperature are relayed to the DC windings, which in turn controls the input power to the furnace windings/elements.
- The major advantage of this controller is that temperature fluctuations are eliminated.

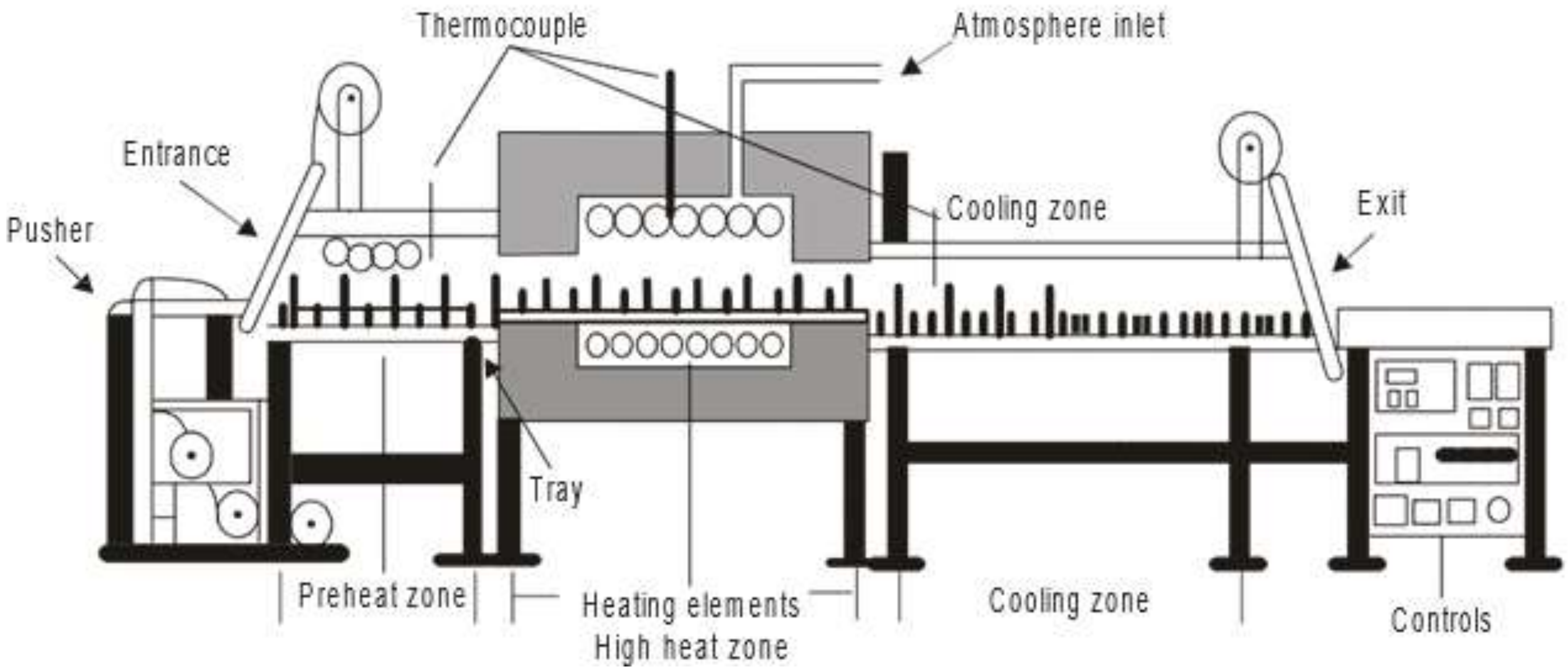
SINTERING FURNACES - classification

- Based on the types of heating - fuel-fired or electrically-heated,
- Horizontally- and vertically-fired fuel fired radiant tube furnaces allow low maintenance operations up to 1,000°C using nickel-chromium alloy tubes and up to 1,100°C with superalloy or ceramic coated tubes.
- Electrically-heated furnaces provide constant temperatures by means of proportional current controllers and the furnaces make use of nickel-chromium (1,150°C), silicon carbide (1,150°C), graphite (2,000°C) and molybdenum (1,850°C), (requires hydrogen atmosphere) as heating elements depending on the temperature

Based on the type of operation

- Continuous or batch type furnaces
- Continuous furnaces are used for large scale production, wherein the green parts are loaded into trays or trolleys which are then passed through the sintering furnace.
- The movement of the trays is controlled so that when the parts come out of the furnace, the sintering is complete.
- Most sintering furnaces are of continuous type.
- In these type of furnaces, movement of the compacts takes place through the various zones sequentially by the use of (a) a conveyor belt system or (b) a roller hearth in which tray loads of green compacts are conveyed by a number of rollers placed parallel to one another or (c) a pusher in which the compacts placed on individual trays are pushed automatically through the three zones for specific times, and (d) walking beam furnaces in which the parts are periodically lifted and

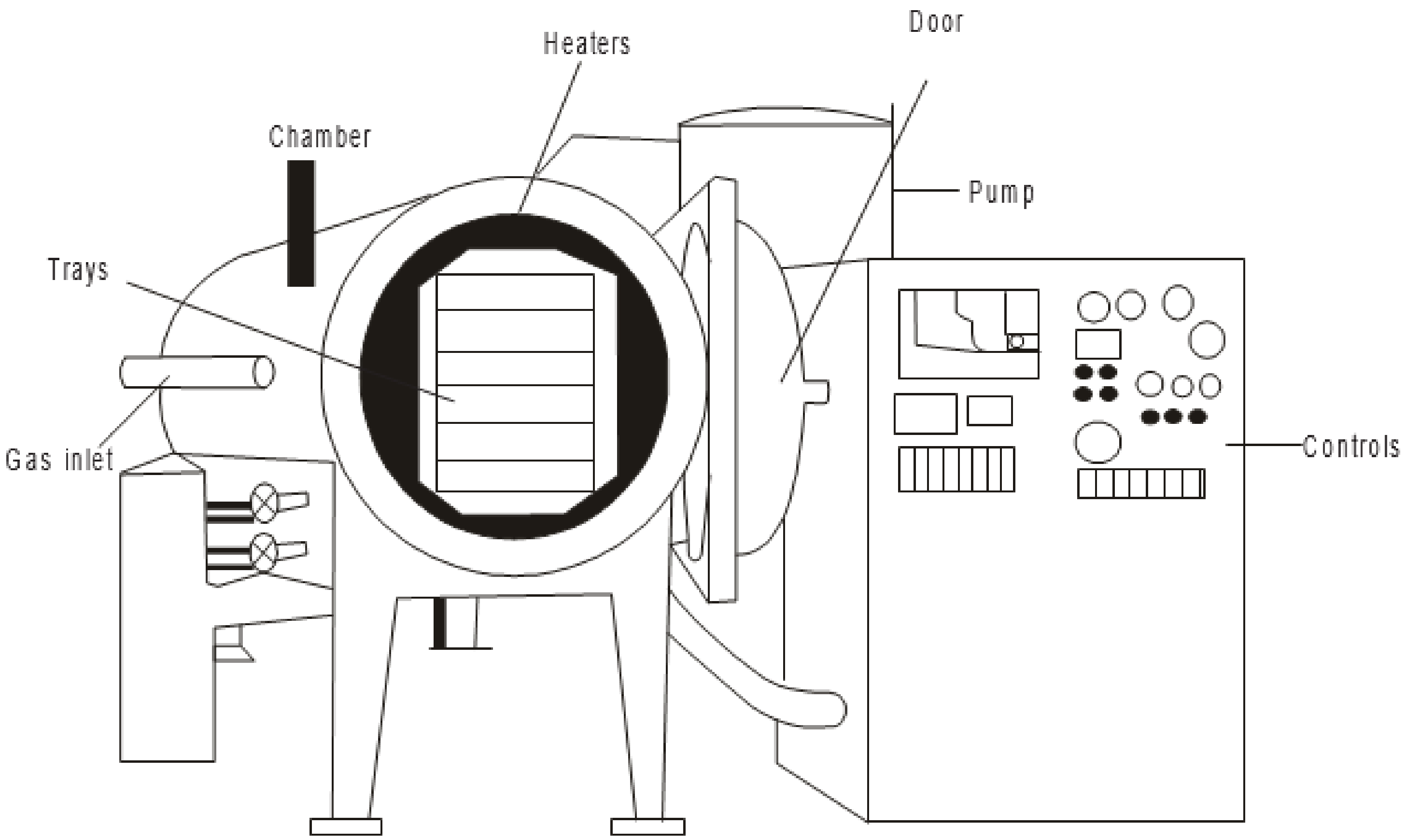
Continuous sintering furnace



- Continuous furnaces are used in the production of porous bearings, sintered carbides as well as structural parts made of steel.
- Continuous furnaces are sometimes provided with muffles when dew point control is very important.
- Muffles are used for isolating products from fuel and the by products of combustion from the heat source.

- Muffles are also important when the products of combustion of gas fired furnaces give off products that are not protective in nature.
- These muffles are made of high heat and oxidation-resistant material like wrought Inconel alloy.
- Mesh belt and roller hearth furnaces are limited by their service temperature (up to 1,150°C), which in turn is dependent on the belt and roller material.
- Walking beam furnaces can operate at temperatures of 1,350°C, because the lifting part is made of a refractory backed by alloy steel.

Batch type sintering furnace



- Batch type furnaces are used when small quantities of parts or parts having different sizes are to be sintered.
- The parts are loaded into trays and pushed through the box type furnace manually.
- Molybdenum or tungsten heating elements heat this type of furnace.
- Molybdenum, tungsten or cemented carbides (WC type) are sintered in this type of furnace.
- Bell type batch furnaces contain a bell, which is lifted-off to load the green parts on the fixed base.
- A retort is placed over the load and is used for circulating a protective gas during sintering.
- The bell then placed in position and is sealed air-tight.

- After sintering is over, both the bell and the retort are taken off and the parts are removed and new parts are charged for the next sintering cycle.

- Bell type batch furnaces with very good atmosphere control have also been used for pressure sintering of friction materials.

Based on the type of atmosphere

- Muffle furnaces, controlled atmosphere furnace or vacuum furnaces.
- Controlled atmosphere furnaces are most commonly used types in sintering.
- These furnaces provide a blanket of a gas (reducing, neutral) which prevents undesirable reactions during sintering.
- The sintering atmosphere must prevent oxidation of the metal surface at the sintering temperature, avoid carburizing, decarburizing and nitriding, must reduce surface oxide films on powder particles, remove absorbed gases and should not contaminate the powder compact.
- The three types of atmospheres prevalent in

- The most widely used sintering atmospheres are reducing type although inert type is used for certain applications.
- Reducing atmospheres can be divided into dry hydrogen or dissociated hydrogen, exothermic atmosphere with low or medium carbon potential and endothermic atmosphere enriched with hydrocarbon gas.
- Inert gas, especially, argon and helium are used only to a limited extent because of their relatively high cost.

- Nitrogen based mixtures also find use in several applications.
- A minimum of 90% of nitrogen prepared by fractional distillation of liquid air mixed with hydrogen and methane is used as an atmosphere replacing endogas.
- These gases are separately pumped in before they are mixed and used in the furnace.
- Along with these gases, by adding 5 to 12% hydrogen a reducing atmosphere is obtained while an addition of CH_4 results in a carburizing atmosphere.

Vacuum sintering

- Expensive is used in special cases like high-temperature sintering of steels alloyed with reactive elements.
- Vacuum sintering furnaces were originally developed for sintering refractory and reactive metal compacts as well as for sintering cemented carbides containing titanium and tantalum carbides.
- These are batch type furnaces in which the compacts to be sintered are placed on graphite trays and heated by high-frequency induction coil or by graphite heating elements.
- The furnaces have provision for lubricant removal, presintering and final sintering.

- Continuous vacuum sintering furnaces with a rotating table allowing compacts to be de-waxed, sintered and cooled are also in use.
- These furnaces are used mainly for sintering of highly reactive metals like titanium, tantalum, zirconium and their alloys, Al-Ni-Co magnets as well as cemented carbides (WC, TiC and TaC grades).
- Presently, these furnaces are employed in the sintering of stainless steel parts also.
- These furnaces are very expensive and require sophisticated equipment and controls.
- Graphite is commonly used as heating element, though tantalum is also used.
- Vacuum furnaces are becoming popular because of the reduced energy requirements compared to

- The above classification is of a general nature and there can be overlapping, for example, vacuum furnaces can be batch or continuous ones and batch furnaces may be controlled atmosphere or muffle furnaces.
- The choice of a suitable furnace atmosphere is of utmost importance in the sintering process.

METALLOGRAPHY OF P/M PARTS

- Metallography of P/M parts is a part of the general metallography. Metallography of P/M materials can be done similar to conventional materials.
- Two distinct aspects in which P/M differs from conventional metallography are

(i) characterization of powder particles

(ii) characterization of porosity.

Characterization of Powders

- Powders are normally characterized for their shape, size as well as their internal structure, (phases and their distribution, porosity and inclusions).
- Powder shape or size determination can be done using SEM.
- TEM is used for obtaining accurate and reliable results, especially for nanosized powders.
- Powder is spread on a conducting medium applied to the specimen holder.
- A conductive coating is applied for ceramic powders before the specimen is observed.

- Care must be taken to avoid particle agglomeration for effective results.
- If the powder is to be characterized for its microstructure, the powders must be mounted, polished and observed after etching with suitable reagents.
- In mounting, care must be exercised to avoid agglomeration of powders.
- Normally optical microscope is used for the study of microstructure.
- Use of electron microscopy is also done sometimes.

Characterization of P/M Parts Containing Porosity

- P/M parts generally contain porosity.
- Materials like sintered bearings contain a significant amount of porosity.
- Metallographic preparation of such a material must be done carefully because polishing can alter the surface details significantly and lead to erroneous interpretation of microstructure.
- Some of the effects of incorrect preparation include:

(i) Partial closing of pores by plastic deformation during grinding,

(ii) Rounding of pore edges, and

Steps to be followed in preparing P/M samples

Sampling

- In case of large parts, small sections are obtained using an abrasive cut-off wheel with water as coolant.
- The cut sections are then washed to remove cutting debris from the pores by thorough rinsing with water.
- Mounting, if necessary, may be done using epoxy resins with the help of a mounting press.

Grinding

- Grinding is carried out on automatic grinding wheels using silicon-carbide paper of 220 grit size and water as coolant.
- Fine grinding is done with 500 and 1,000 grit size.
- A speed of 300 rpm with a load of 90, 60 and 30 N is used for the three grinding steps.
- The specimens are then washed using ethanol in an ultrasonic cleaning unit.
- Lapping is carried out on automatic wheels in three steps. In the first step the load is 90 N, in the second it is 60 N, and thereafter 30 N.
- Spray containing 6 micron diamond particles is applied.
- Polishing is carried out using a speed of 150 rpm, and the total lapping time, depending on the material, is between 5 and 15 minutes.
- After lapping, ultrasonic cleaning in ethyl alcohol is

Impregnation

- This is done to preserve the pore edges as well as for the prevention of pore closure during polishing and is usually carried out with epoxy resin or molten metal.
- Before impregnation the sample must be dried in vacuum to remove the moisture, especially within the pores.
- This helps to prevent the oozing out of the impregnated material during etching and observation.
- Impregnation is then carried out, also under vacuum.

- Regrinding may be done using 500 and 1000 grit

Final Polishing

- This is carried out with fine (particles with 6, 3 and 1 micron diameter) diamond polishing spray using a load of IN for one minute.
- For the final polishing step, an alkaline suspension diluted with distilled water is used.
- The specimens are thoroughly rinsed with water, dried and taken for metallographic observation.

Etching

- Etching is generally performed by immersion.
- Etching reveals these important features: grain size, and the presence of different phases including non-metallic inclusions.

- Same etchant used for the wrought product is used for P/M materials also.